5.14 Construction of a Rhombus

Let us now construct a rhombus with the given measurements

(i) One side and one diagonal

(ii) One side and one angle

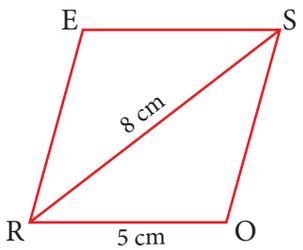
(iii) Two diagonals

(iv) One diagonal and one angle

5.14.1 Construction of a rhombus when one side and one diagonal are given

Example 5.34

Construct a rhombus ROSE with RO = 5 cm and RS = 8 cm. Also find its area.



Rough diagram

Solution:

Given: RO = 5 cm and RS = 8 cm

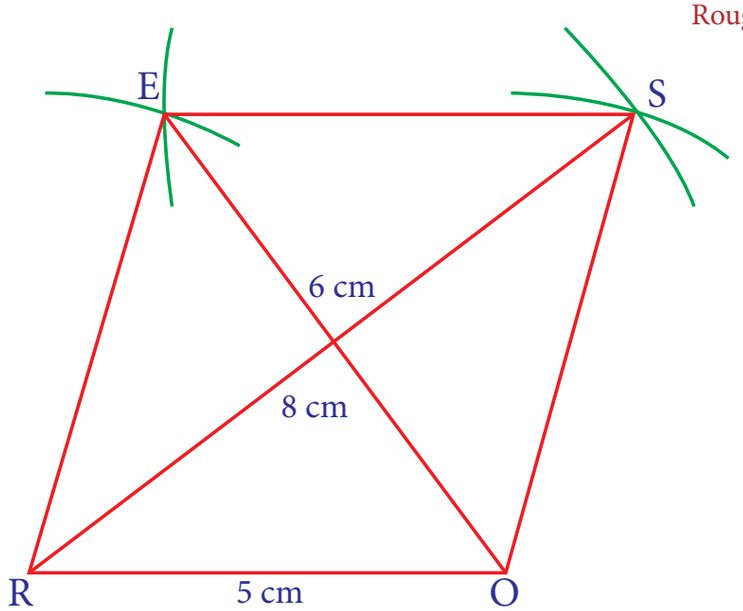


Fig. 5.53

Steps:

1. Draw a line segment RO = 5 cm.
2. With R and O as centres, draw arcs of radii 8 cm and 5 cm respectively and let them cut at S.
3. Join RS and OS.
4. With R and S as centres, draw arcs of radius 5 cm each and let them cut at E.
5. Join RE and SE.
6. ROSE is the required rhombus.

Calculation of Area:

Area of rhombus ROSE $= \frac{1}{2} \times d_1 \times d_2 \; sq.units$

$= \frac{1}{2} \times 8 \times 6 = 24 sq.cm$

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